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## **AI-Driven Vessel Idle Management: Retrofittable Start-Stop System For Emissions Reduction And Operational Optimization**

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### **Abstract**

This paper presents a scalable, AI-integrated technology to reduce fuel consumption and carbon emissions in offshore vessel operations through automated idle management. Idle management enables retrofitting existing diesel-driven propulsion and auxiliary engines with an intelligent start-stop system. The scope includes integration into offshore logistics fleets, operational impact evaluation, and system performance validation in real-world oil & gas environments.

Idle Management uses lithium-ion battery technology and AI-based control logic to manage vessel engine idle periods dynamically. It is deployed in the three retrofit phases

- Installation on propulsion engines for automated standby/restart.
- Hybridization with auxiliary systems including AC/DC conversion and battery integration.
- Shore power connectivity enabling full engine shutdown during port stays.

The system is complemented by integration with a telematics and data platform for live performance monitoring. The methodology involves field trials on offshore support vessels (OSVs), tracking engine runtime, fuel consumption, carbon dioxide equivalent CO<sub>2e</sub> output, and operational efficiency metrics using real-time telematics data.

Initial deployments on OSVs operating under unsteady load profiles demonstrated significant reductions in idle time and fuel use. One use case showed a reduction of 7,200 engine hours annually, saving 250,000 liters of, equating to 700 metric tons of CO<sub>2e</sub> avoided per vessel. The automation of standard operating procedures (SOPs) minimized human error and reduced operator workload, transitioning them to supervisory roles. Idle management also improved engine longevity by reducing cold starts and unnecessary runtime. Additionally, integration with existing control systems allows performance validation, supporting compliance with environmental regulations and customer-specific fuel reduction goals. Results indicated an increase in operational uptime (WT, PT), reduced downtime (DT), and enhanced asset availability. The full implementation of all three retrofit phases enables vessels to transition toward hybrid operation and eventual full electrification, aligning with Net Zero targets. The modularity and ease of integration of idle management makes it suitable for large-scale adoption across the offshore logistics sector.

This paper introduces a commercially viable, retrofit-ready technology combining AI control, battery systems, and automated SOP execution to tackle fuel and emissions inefficiencies in marine operations. Unlike conventional idle reduction methods, idle management delivers measurable, verifiable benefits through telematics integration. It provides a pathway for decarbonizing existing offshore assets without full equipment replacement—bridging current operations with future hybrid and zero-emission standards.

## Introduction

Offshore Supply Vessels (OSVs), especially Platform Supply Vessels (PSVs), play a central role in the logistics chain of offshore oil production. They move in cyclical routes between oil ports and offshore platforms. Due to highly specialized drives, long idle times occur. For example, during loading and unloading, waiting times and stationary positioning.

Ships in offshore use typically have three operating states:

- (Full) Ride at (100%) load
- Offshore operation with changing load
- Port operation (loading and unloading, auxiliary energy)

In states 2 and 3, the main or auxiliary engines responsible for propulsion of the ship run at idle, although no propulsion power is required. In contrast, auxiliary units usually do not run at the ideal operating point when driving between the oil port and the platform.

- excessive idle hours,
- high maintenance requirements,
- unnecessary fuel consumption,
- increased emissions,
- reduced plant availability.

A significant proportion of the engine's running time is due to low-capacity or idling conditions - with negative effects on fuel consumption, emissions, maintenance and service life.

This paper presents an approach to how intelligent control can be used to optimize engine operation on these ships. This is done by means of an AI-supported retro-fit system for cross-engine control and idle management using the Engine Standby Controller (ESC). This is a system to make diesel-powered large engines dynamically controllable. It detects idling phases, optimizes the operating strategy of individual engines and thus reduces emissions, operating hours and wear. The solution has a modular structure and is implemented in several steps.

## Ship architecture and engine configuration

The reference vessel is a medium-sized PSV (~80 m in length). Platform Supply Vessels (PSVs) are specialized supply vessels that play a central role in offshore oil and gas logistics. They transport equipment, supplies, fresh water, chemicals and other supplies from land bases to offshore platforms. In addition, they provide support during crew changes, emergency operations and the removal of waste or return loads. Due to their high maneuverability, often supported by dynamic positioning systems (DP2 or DP3), PSVs can operate safely on offshore structures even in difficult sea conditions. Their operation is highly cyclical - with frequent changes between driving, idle times and loading and unloading processes.

The ship has a diesel-electric propulsion architecture with a total installed capacity of around 7,200 kW. The main drive is provided by two Caterpillar 3512C diesel generators (ME1 and ME2), each with an output of about 1,800 kW, whereby only ME1 is active in normal operation and ME2 is available as redundancy. Two additional machines (BE1 and BE2), each with 1,500-1,800 kW, are available to operate the dynamic positioning system (DP2), which are switched on to the platform during stationary operations. The on-board power supply is ensured by two auxiliary units (AE1 and AE2): AE1 continuously supplies 150 kW for the hotel and system loads, while AE2 supplies cranes and pumps with 100 kW of power when required and is otherwise kept idle as immediate redundancy.

The energy supply is fully diesel-electric; the ship has azimuth propulsion and is DP2 classified. Ships with DP2 (Dynamic Positioning Class 2) classification have a redundant system for automatic positioning that remains functional in the event of a single failure, such as a generator, control unit or propulsion system. In contrast, DP1 ships have no redundancy; a single mistake can lead directly to the loss of position. DP3 systems go beyond DP2 and additionally require physical separation of all critical systems to ensure safe positioning even in the event of fire or flooding. For Platform Supply Vessels (PSVs) that regularly operate close to offshore platforms, DP2 offers the necessary safety standard while at the same time providing reasonable technical and economic effort.

The ship under consideration here is operated in the Arabian Gulf

The typical operating profile of the vessel includes 10 to 18 shuttle trips a day between the Abu Dhabi terminal and an offshore platform about 10 kilometers away. Each route takes around 15 to 18 minutes at a cruising speed of 10 to 12 knots. During the platform stay of an average of 15 to 25 minutes, the vessel is kept stable in dynamic positioning mode (DP2), with both backup engines (BE1 and BE2) active. In the port, goods, liquids or equipment are handled within 10 to 20 minutes using AE1 and AE2 auxiliary units for cranes and pumps. At night (22:00-06:00), the ship is in idle mode in port, with AE1 taking over the base load throughout and AE2 remaining operational at idle.

## Idle Management with Engine Standby Controller

The ESC is an intelligent start-stop module with an integrated battery system that can manage the diesel operation of each individual engine on a situational and autonomous basis. The central technical basis is a powerful 6 kWh lithium-ion battery (25.6 V), which is responsible for cold starts, maintaining hydraulic pressure and preheating the machines. The system detects when the engine is not needed, switches it off in a controlled manner and keeps it in a ready-to-use state. As soon as there is a power requirement again - e.g. due to a change in driving condition or requirements from the on-board electrical system - the engine automatically restarts without manual intervention by the operator. Active conditioning in Ready mode allows restart within one second.

The ESC takes over the entire transition logic between the following operating modes:

- **Engine Running:** Normal operation underload
- **Engine Idle:** Operation without performance requirements
- **Engine Off / Ready:** ESC has shut down the engine, but operational readiness remains
- **Automated Restart:** Load request detected, ESC activates active heating and lubrication, engine starts

The starting process is not only electrified (by starter motors), but also actively supported by:

- Automatic lubrication circuit activation (Active Lubrication),
- Integrated preheating (VULCAN Heater), if required ( $< 0^{\circ}\text{C}$ )

- battery-supported on-board power supply (incl. intelligent charging circuit with redundancy),
- Automatic diagnostics and self-monitoring (incl. remote interfaces).

The ESC hardware is robustly designed for marine use (IP65, -30 °C to +45 °C), completely maintenance-free and scalable. The system can be operated either directly on the main machine, on auxiliary units or in a redundant configuration with several ESCs. In combination with battery buffer storage systems, hybrid converters (AC/DC) or shore power connections, the ESC forms the basis for modular electrification of maritime propulsion systems.

A decisive aspect is the fully automated control: the ESC does not require any manual intervention. The operator no longer has to actively start or stop the engine - the entire idle optimization is software-controlled, based on operating parameters, sensor values and, if necessary, external load data (e.g. via telematics systems such as a telematics platform). This reduces human error, lowers workload and increases safety.

## **Ship performance control by means of cross-engine map optimization**

The control system is basically decentralized.

Each of the engines used on the ship is equipped with an idle management system. These systems have independent control units. The start-stop functionality is therefore also available locally and stand-alone. This increases the resilience of the system. All components are networked with each other and transmit the data to the central control unit via a hybrid cloud-based platform.

The actual engine fleet intelligence results from the sum of the individual controllers with the corresponding actuators but is implemented by means of central KPI specifications. This includes the AI functions of the setup. The operating parameters of the individual units are evaluated across the board. In this way, the map is recorded and optimized.

## **Microgrid topology**

The electrified or electrifiable components can be connected via a decentralized 1,000 / 600 V DC and/or 480 V AC microgrid. On the generation side, the microgrid in this case consists of the motors already existing on the ship. The use of a microgrid enables the exchange of energy between the aggregates and at the same time ensures increased redundancy. The connection is made via mobile, easily configurable plug connections. A central control system is not required; instead, all units act cooperatively via networked control loops.

## **Adaptive map detection with AI**

The vessels under consideration operate in dynamic environments with highly volatile load profiles, irregular downtimes and constantly changing operating conditions. For example, the azimuth drives have to react to highly volatile flow changes. In such scenarios, manual optimization of engine operation is neither reliable nor efficient. The aim of AI-based map recognition is therefore to create an automated, adaptive decision-making unit that selects the best operating point in real time in terms of energy and operation.

Previous control concepts rely on predefined static characteristic curves or operator intelligence. The latter often leads to suboptimal behavior - for example, due to unnecessary revving, inefficient half-load behavior or delayed shutdown. Adaptive map recognition replaces these procedures with continuously learning logic that is tailored to the actual operating data. This significantly reduces consumption, emissions and wear.

## Machine Learning Model

The machine learning model (ML model) is an individual model for each engine setup. It is created from the combination of the engines used on the ship.

The ML model for map recognition is integrated directly into the individual ECUs, such as that of the ESC (Engine Standby Controller). It is based on supervised learning methods that work with multivariate input data such as speed ( $n$ ), outlet pressure ( $p$ ), coolant temperature ( $T$ ), torque ( $M$ ) and energy demand ( $E$ ). The target is the specific fuel consumption [g/kWh].

The farm map is mapped as a multidimensional farm point map. In pilot tests, grids with up to 10,000 points ( $100 \times 100$ ) were used. The initial training phase took place offline on historical operating data, divided into representative clusters: start-ups, idle, constant operation, partial load operation, high load with transients.

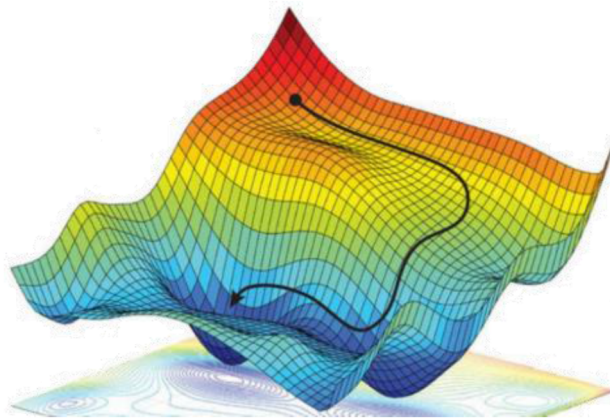


Figure 1—Multidimensional Farm Point Map

Continuous online learning takes place during real-time operation. The model is dynamically updated to compensate for drift effects (e.g. due to component aging, weather or load shifts). The weighting is done via a Bayesian update procedure that links priority distributions with new data.

### Online adaptation in the field

The AI system works completely autonomously and locally. Each unit learns from its own operating history. The operating parameters are continuously evaluated via the integrated SOPHIA cloud, and an overarching strategy specification is issued. The strategy specification can be expanded to include, for example, optimization for maximum efficiency, minimum emissions or longest service life. The local ESC unit receives this input and adapts the prioritization logic accordingly.

The online adaptation reacts to fluctuations such as:

- Water temperature (influence on cooling limits)
- Time of day (start/stop behavior at changing temperatures)
- Frac stage (pressure build-up, peak flow, holding phase)
- Aggregate state (availability assessment via digital twin)

Like intelligent cruise control in the automotive sector, AI regulates the optimal pump speed and the corresponding pressure range in such a way that fuel consumption is within the optimal range. There is no need for faulty operator intervention or delayed reactions. The result: +12% efficiency, -18% emissions (measured in pilot projects with  $4 \times$  ESC pumps cumulative over 1,600 h).



## Architecture

The control hardware of the individual actors is based on an ARM Cortex-M7 with a dedicated FPU (Floating Point Unit). A RAM buffer of 512 kB is available for the real-time model, which is queried cyclically by the ESC rules software. The ML model is modular and can be updated by OTA update via SOPHIA.

Communication is done via MQTT with SOPHIA, internally via J1939-CAN. In the event of a communication failure, the ML system continues to operate locally. For fleet operation, the architecture is designed in such a way that learning effects from other units can also be aggregated ("transfer learning").

## System integration on the PSV

As part of the system integration of the Engine Standby Controller (ESC) on a PSV, a three-stage concept based on real operating conditions was implemented. The aim is the gradual transformation of existing systems into automated, highly efficient energy management with a focus on emission and fuel savings.

### Stage 1: ESC on the main propulsion engine

In the first expansion stage, the ESC will be integrated directly into the control system of the main machine. As soon as there is no power requirement, the system automatically puts the engine into a "ready mode" - the engine is switched off, but ready to start at any time. Power requirements, for example via the propeller or navigation control, are detected by the system itself. The engine is then started automatically, including all necessary preheating and lubrication cycles. There is no need for manual control by the on-board crew - the operator becomes the supervisor, and the ESC fully executes the operating logic. In addition, there is a connection to telematics, which enables automatic operating hours recording. Manual logbook entries are eliminated, and human errors are minimized. These measures can reduce the idle time of the main engine by over 70%, resulting in reduced diesel consumption, lower maintenance costs and unchanged responsiveness.

### Stage 2: ESC on auxiliary units + integration of a battery system

The second expansion stage addresses the auxiliary machines. A second ESC is coupled to the genset, supplemented by a battery system (Li-Ion, 6 kWh), which is integrated into the on-board electrical system via a hybrid AC/DC converter. In idle phases, the battery takes over the basic supply. The auxiliary machine is switched off completely and only activated automatically when necessary. The system thus not only optimizes the energy consumption of both machines, but also smoothes load profiles and prevents unnecessary starts. In addition, the basis for the later integration of renewable energy sources is created, for example through photovoltaics or fuel cells.

In the final expansion stage, a shore connection module will be integrated to ensure emission-free operation in ports or offshore. The ESC system coordinates the automatic switchover to external power supply, so that no machine must be operated anymore. The measure eliminates all idling emissions in the port area, meets future requirements of low emission zones and completes the ship's holistic energy optimization.

## AI-powered operating point optimization

Each ESC unit on the ship contains an embedded machine learning module for precise control of the respective operating point. This module follows a hybrid control approach: It combines data-based AI optimization with hard-coded security and system boundaries. The control system is based on continuously collected input data such as motor speed, torque, electrical load, battery voltage and ambient temperatures. The model's target is to minimize specific fuel consumption in grams per kilowatt hour (g/kWh).

The ESC AI analyzes over 10,000 multidimensional load points from real operation and continuously adapts its model during operation. This allows the system to detect and compensate for drift effects caused by aging, temperature or external conditions. The overarching control strategy is to detect idle phases with

an expected duration of more than three minutes at an early stage and to put the engine into a controlled "soft shutdown" mode. If power is required again, the engine starts automatically - including preheating and lubricant circulation - within a few seconds.

During operation, the model also detects transitions to critical operating states, such as switching to dynamic positioning mode (DP2), and initiates preventive measures, such as switching on the second backup motor (BE2) to ensure redundancy. In addition, the system prioritizes safety-critical consumers such as UPSs, navigation systems or DP expansion joints and preferentially ensures their power requirements.

The "AI-in-the-loop" architecture ensures that all control variables proposed by the AI are validated within a deterministic framework. For example, if engine temperatures are exceeded, rapid load changes or low battery levels are automatically switched to more conservative control strategies. This hybrid structure enables data-driven optimization without compromising security or redundancy requirements.

All communication between the ESC units is done via standardized industrial protocols such as CAN (J1939) and MQTT. EKU's own cloud platform SOPHIA can be used to distribute model adaptations centrally, synchronize fleet strategies and maintain digital twins for each drive. This makes the system scalable for both individual vessels and fleet applications and allows learning transfer between units with similar load profiles.

## Simulation and evaluation

For the quantitative evaluation of the system effect, a high-resolution simulation was carried out based on realistic operating profiles. The underlying analysis included a full month's operation (July 2025) of a typical offshore supply vessel with daily shuttle traffic between an oil port in Abu Dhabi and an offshore platform about 10 kilometers away.

The simulation model worked with a temporal resolution of one minute and depicted 10 to 18 operating cycles per day. Transit phases, layover times at the platform with dynamic positioning, port handling processes and nightly rest periods were taken into account. In addition to the physical parameters of the on-board electrical system and the machines (speed, power, status), load flows in energy storage systems and start-stop processes of the individual units were also recorded. In total, the generated dataset comprised more than 44,000 individual rows with seven observed variables for six engines.

The evaluation showed an average utilization of 34% for the primary main engine ME1, with an idle rate of only 2% and a switched off state of over 64% of the total time - made possible by the intelligent intervention of the ESC. The backup engines BE1 and BE2 were primarily used during dynamic positioning and had a combined activity duration of about 20%. The on-board electrical system was largely supplied by AE1, whose output was in the range of 135 to 200 kW. AE2 was activated as an additional redundancy for cranes and pumps in 6 to 9 missions per day but remained idle most of the time.

These results make it clear that the use of the ESC not only achieves significant operating time savings for individual units, but also controls the overall system utilization more transparently and efficiently. In combination with battery systems or a shore power connection, these effects can be further amplified.

## Energy & Emissions Balances

The simulation results were then used to quantify the energy and environmental impact of the ESC system. The focus was in particular on reducing idle times, fuel consumption, CO<sub>2</sub> emissions and the technical oil requirements of the engines.

By implementing the Engine Standby Controller, the idle time of the primary main motor (ME1) was reduced by 54% compared to conventional operations. In the case of the BE1 and BE2 backup engines, which are mainly used in DP2 operation, the no-load reduction was around 40%. These reductions led to a significant decrease in total diesel consumption over the year of operation.

In concrete terms, this results in annual savings of around 51,000 liters of diesel fuel per ship. With an emission factor of around 2.68 kg CO<sub>2</sub> per liter of diesel fuel in accordance with the IPCC Directive, this corresponds to a CO<sub>2</sub>e reduction of around 171 tons per year and ship. In addition, lubricating oil consumption is significantly reduced due to the reduced engine running time - in the simulation under consideration, a saving of around 830 liters of engine oil per year was determined.

These figures impressively demonstrate that the ESC system not only brings economic advantages but also makes a tangible contribution to reducing emissions in maritime offshore logistics. In combination with other measures such as energy storage or shore power use, this effect can be further intensified.

## Maintenance, cost-effectiveness and regulatory embedding

In addition to the environmental and operational benefits, the implementation of the ESC also brings significant improvements in terms of maintenance and service life of the machines. By deliberately avoiding unnecessary cold starts - one of the main causes of premature wear - a reduction of these starts of over 70% could be demonstrated in the simulation. As a result, the mechanical stress on the main machines is significantly reduced, resulting in an extended service life of 12% on average. At the same time, the intelligent start-stop strategy enables maintenance intervals to be extended and reduces predictable downtimes.

There are also clear economic benefits: the expected payback period for the ESC system is less than 24 months (based on phase 1). The annual savings per PSV amount to around USD 400,000, which represents a potential savings of over USD 8 million per year for a fleet of 20 vessels - such as those operated by ADNOC Marine Operations.

The system architecture is fully compatible with existing regulations. Relevant standards such as IMO MSC/Circ.645 for position redundancy, DNV-ST-0111 regarding DP2 conformity (changeover time < 45 s) and the IMCA guideline M 182 are complied with by ESC-compliant load management.

## Data validation, emissions modelling and technical comparison

An integral part of the system is the continuous validation and analysis of the operating data. ESC provides high-frequency telemetry data, including engine speed, load conditions, start-stop cycles and fuel consumption. These are supplemented by a telematics platform that records sailing histories, DP times, port handling and layover times. In a common KPI dashboard, metrics such as idle rates, dispersion of operating points or DP redundancy assessments can be derived from this. In addition, the ESC backend detects anomalies in operating behavior and can thus react to deviations at an early stage.

The emission modelling is based on the measured diesel consumption, which is converted into CO<sub>2</sub>e by means of a standardized emission factor (IPCC: ~2.68 kg CO<sub>2</sub> per liter of diesel). Other emissions such as nitrogen oxides (NO<sub>x</sub>) and carbon monoxide (CO) are interpolated based on manufacturer characteristics (e.g. Caterpillar 3512 Marine). Thanks to optimized operating points, the ESC not only reduces consumption, but also specific emissions per kilowatt hour. Phase 2 scenarios with energy storage and shore power connection also enable emission-free port operating time ("zero emission berthing").

The comparison with conventional operation (SOP - Standard Operating Procedure) shows the clear superiority of the ESC approach:

Table 1—Metrics

| Metric             | SOP operation | ESC Phase 1 | ESC Phase 2 |
|--------------------|---------------|-------------|-------------|
| 0 idle time        | 32%           | 15%         | < 5 %       |
| Diesel consumption | 100 %         | 85%         | 70 %        |



| Metric                    | SOP operation  | ESC Phase 1 | ESC Phase 2               |
|---------------------------|----------------|-------------|---------------------------|
| CO <sub>2</sub> emissions | 100 %          | 83%         | 65 %                      |
| Start-stop cycles         | manual         | automatic   | intelligent (with memory) |
| Maintenance               | unquestionable | prolonged   | State-based               |

The economic feasibility study in detail shows that the investment costs per ship for pure ESC installations are around USD 120,000. From phase 2 onwards, a return on investment (ROI) of less than six months is achieved. In addition to the direct savings, the reduced maintenance requirements and higher availability also contribute to cost-effectiveness. In addition, there are positive effects from emission certificates and optimized billing of port and layover times via a telematics platform.

## Conclusion and outlook

The Engine Standby Controller (ESC) from EKV Power Drives enables the gradual decarbonization and digitalization of existing ship fleets. By combining adaptive engine control, adaptive AI logic and cloud-based analytics platform, significant savings in fuel, emissions and maintenance costs can be achieved - with full compatibility with DP2 requirements and existing operating processes. The combination with a telematics platform also creates a reliable database for regulatory, operational and economic optimization. The presented multi-stage rollout approach presents a transferable model for similar fleets worldwide.